

Ideation Phase

Brainstorm & Idea Prioritization Template

Date	12 June 2026	Team ID	N/A (Solo Project)
Project Name	Network Anomaly Detection	Maximum Marks	4 Marks

Brainstorm & Idea Prioritization Template:

Brainstorming provides a free and open environment that encourages everyone within a team to participate in the creative thinking process that leads to problem solving. Prioritizing volume over value, out-of-the-box ideas are welcome and built upon, and all participants are encouraged to collaborate, helping each other develop a rich amount of creative solutions.

For this solo project, brainstorming was conducted as a structured self-ideation exercise: candidate approaches were listed, grouped by theme, and prioritised on an importance-vs-feasibility grid to arrive at the final solution selection.

Reference: <https://www.mural.co/templates/empathy-map-canvas>

Step-1: Team Gathering, Collaboration and Select the Problem Statement

Team Gathering	Problem Focus Area	Define Problem Statement (HMW)
Solo Developer: Sree Sai MIT Manipal — School of Computer Engineering Role: Full-stack ML developer SmartBridge / IBM Cyber Security Analyst Course (SWAYAM Plus) Brainstorming method: structured self-ideation — listing candidate approaches, evaluating trade-offs, selecting the best fit for solo scope.	Primary concern: Modern networks generate millions of flows per day. Embedded within that volume are attacks — DoS, port scans, credential-stuffing, stealthy infiltrations. Core constraint: no labelled attack data should be required during training (unsupervised). Solo scope: must run on free-tier CPU, complete in < 2 minutes, and produce explainable output.	How Might We... "How might we automatically detect anomalous network traffic flows without requiring any labelled attack data during training, so that the system can flag unknown and zero-day threats with minimal human intervention — and produce results that a security analyst can interpret and act upon?"

Key rules applied during brainstorming: Stay on topic. Defer judgment. Go for volume. Encourage out-of-the-box ideas. If possible, be visual.

Step-2: Brainstorm, Idea Listing and Grouping

2A — Raw Idea Listing (Sticky Notes)

All ideas generated during the self-ideation session, before any filtering or judgement:

Algorithms	Isolation Forest (sklearn) Unsupervised; fast	k-Means Clustering Simple; no score	Autoencoder (Deep Learning) High accuracy	One-Class SVM Good for novelty Slow on large data	Local Outlier Factor (LOF) Density-based	
Datasets	CICIDS 2017 78 features 14 attack types	NSL-KDD Classic benchmark Older data	KDD Cup 99 Well-known Not realistic	UNSW-NB15 Modern attacks Less documentation	Live packet capture (Scapy) Complex setup	
Features	Use all 78 raw features Complete info	Select top correlated only via EDA	PCA for dimensionality reduction	Packet length stats only Fast & simple	Temporal / sliding window features	

Output	Binary label Normal / Anomaly per flow	Continuous anomaly score for severity tiers	Per-attack-type detection rate breakdown	Results CSV export for analyst review	Live REST API streaming alerts to SIEM	
Tools	Python + scikit-learn Primary stack	Google Colab free-tier CPU Zero install	TensorFlow / Keras for deep learning models	Matplotlib + Seaborn for dashboard	joblib for model serialisation & reuse	

2B — Idea Grouping (Clustered by Theme)

Ideas grouped into five clusters, each given a cluster label:

Cluster	Ideas in Cluster	Cluster Label / Decision
Cluster A	Isolation Forest · k-Means Clustering · Autoencoder (Deep Learning) · One-Class SVM · Local Outlier Factor	Detection Algorithm Selection → Isolation Forest selected: unsupervised, scales to 2.1M rows, trains in ~20s, interpretable anomaly score, no labelled attack data required.
Cluster B	CICIDS 2017 · NSL-KDD · KDD Cup 99 · UNSW-NB15 · Live packet capture (Scapy)	Dataset Selection → CICIDS 2017 selected: 78 features, 14 realistic attack types, publicly available, pre-processed CSV (no raw .pcap parsing needed).
Cluster C	All 78 raw features · Top correlated only · PCA reduction · Packet length stats only · Sliding window features	Feature Engineering Strategy → All 78 features used (EDA confirms breadth is valid); StandardScaler applied to equalise ranges. EDA-identified top features noted for future tuning.
Cluster D	Binary Normal/Anomaly label · Continuous anomaly score · Per-attack detection rate · Results CSV export · Live REST API / SIEM alerts	Output Design → Both binary label AND continuous score produced (score_samples()). CSV export for analyst review. Per-attack-type detection rate included in dashboard. Live API deferred to future scope.
Cluster E	Python + scikit-learn · Google Colab (free-tier) · TensorFlow/Keras · Matplotlib + Seaborn · joblib serialisation	Technology Stack → Python + scikit-learn + Google Colab selected: zero-install, sufficient free-tier compute, all needed libraries available. TensorFlow/Keras excluded (GPU required, overkill for tabular data).

Step-3: Idea Prioritization

Each candidate idea was placed on an Importance vs Feasibility grid to determine which ideas were both important to the project's goals and feasible within solo academic constraints.

	HIGH FEASIBILITY (Low cost / effort / complexity)	MEDIUM FEASIBILITY (Moderate effort / complexity)	LOW FEASIBILITY (High effort / requires infrastructure)
HIGH IMPORTANCE	★Isolation Forest (sklearn) ★CICIDS 2017 dataset ★All 78 features + StandardScaler ★Continuous anomaly score ★Binary Normal/Anomaly label ★Results CSV export ★Google Colab + Python ★joblib model serialisation ★Matplotlib/Seaborn dashboard	● Top-correlated features only ● Per-attack-type detection rate ● Hyperparameter tuning (contamination from EDA)	○ Ensemble model (IF + One-Class SVM) ○ Live REST API / SIEM integration ○ Real-time streaming inference
MEDIUM IMPORTANCE	● PCA dimensionality reduction ● NSL-KDD (alternative dataset)	○ Local Outlier Factor (LOF) ○ k-Means Clustering ○ Autoencoder (Deep Learning)	✗TensorFlow/Keras autoencoder ✗Sliding window temporal features ✗Live packet capture (Scapy)
LOW IMPORTANCE	○ KDD Cup 99 (legacy dataset)	✗One-Class SVM (quadratic scaling)	✗UNSW-NB15 (less documentation) ✗Docker containerisation (MVP scope)

★	Implemented in this project	●	Partially implemented / informing design	○ / ✗	Deferred to future scope or excluded	
---	-----------------------------	---	--	-------	--------------------------------------	--

Prioritization Outcome — Selected Solution

Decision Area	Selected Approach & Justification
Algorithm	Isolation Forest — highest importance AND highest feasibility: unsupervised, no GPU required, trains in ~20 seconds on 2.1M rows, produces interpretable anomaly scores.
Dataset	CICIDS 2017 — only modern realistic benchmark with 78 pre-engineered features and 14 attack types available as pre-processed CSV, eliminating the need for raw packet parsing.
Features	All 78 features with StandardScaler normalisation. EDA correlation analysis (Section 3, Milestone 3) used to identify top predictors for documentation and future feature selection work.
Hyperparameters	contamination=0.17 (derived empirically from EDA's 16.9% attack ratio rather than using the default 0.1), n_estimators=100, random_state=42, n_jobs=-1.
Output	Binary label (Normal/Anomaly) + continuous anomaly score (score_samples()) + per-attack-type detection rate table + four-panel Matplotlib/Seaborn dashboard + anomaly_results.csv export.
Tech Stack	Python 3.x + Pandas + NumPy + scikit-learn + Matplotlib + Seaborn + joblib, running in Google Colaboratory (free-tier CPU). No GPU, no cloud database, no server infrastructure required.
Deferred / Excluded	Autoencoder/deep learning (GPU dependency, overkill for tabular data), One-Class SVM (quadratic scaling unworkable at 2.1M rows), live streaming API (out of solo academic scope), Docker deployment (future scope).